Lesson 5: Introduction to EDA

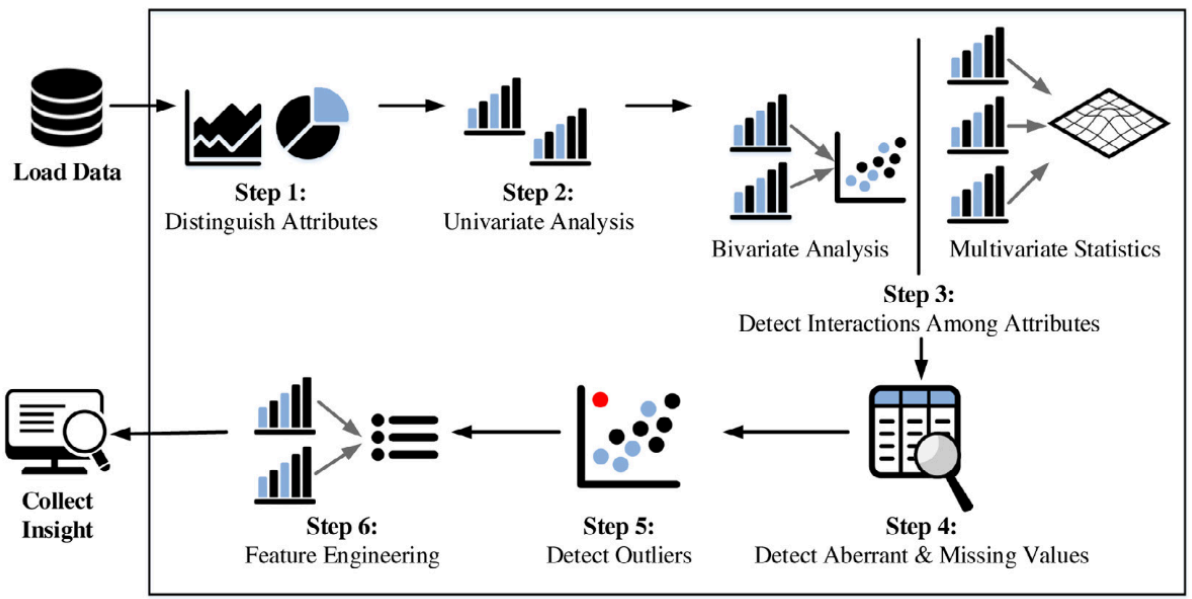
Introduktion til Sundhedsdata

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2026-09-25

What is exploratory data analysis?

- Build knowledge / intuition about your dataset by visualizing it
- Essential for troubleshooting
- Prerequisite for analysis



Purpose of EDA

- Informs modeling choice and handling of variables
- Important to have enough background so that you can validate and troubleshoot analysis
- Often the first thing a reader will want to see

Best practices for EDA

- Done **before** analysis starts
- Built-in to data management
- Fact-checking your intuition and pipeline:
  - Are time trends in a variable stable over time, or do we see unexpected jumps?
  - Is there missingness in this data?
  - Do variables fall within an expected range?

What makes a good EDA plot?

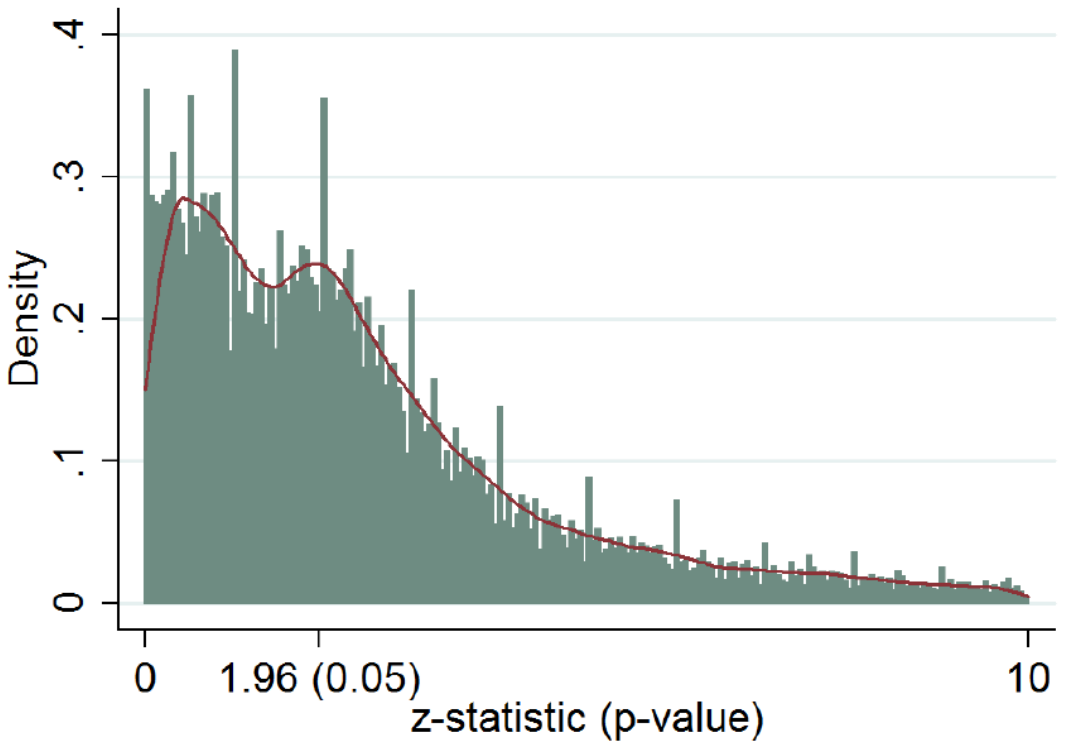
- A figure answers one question at a time
- Someone should be able to understand the figure without you explaining it:
  - Descriptive title and axes legends
  - Clear description of the data source, population represented
- In line with good practices for data visualization

Where to look things up

- **R Graph Gallery** — many different chart types with R code: `r-graph-gallery.com`
- **Kabacoff, *Data Visualization with R***
- **ggplot2 cheat sheet**
- Examples of great data visualization:
  - The Economist's *Graphic Detail* — `economist.com/graphic-detail`
  - *Our World in Data* — `ourworldindata.org`

Univariate EDA

Example: p-values in publication



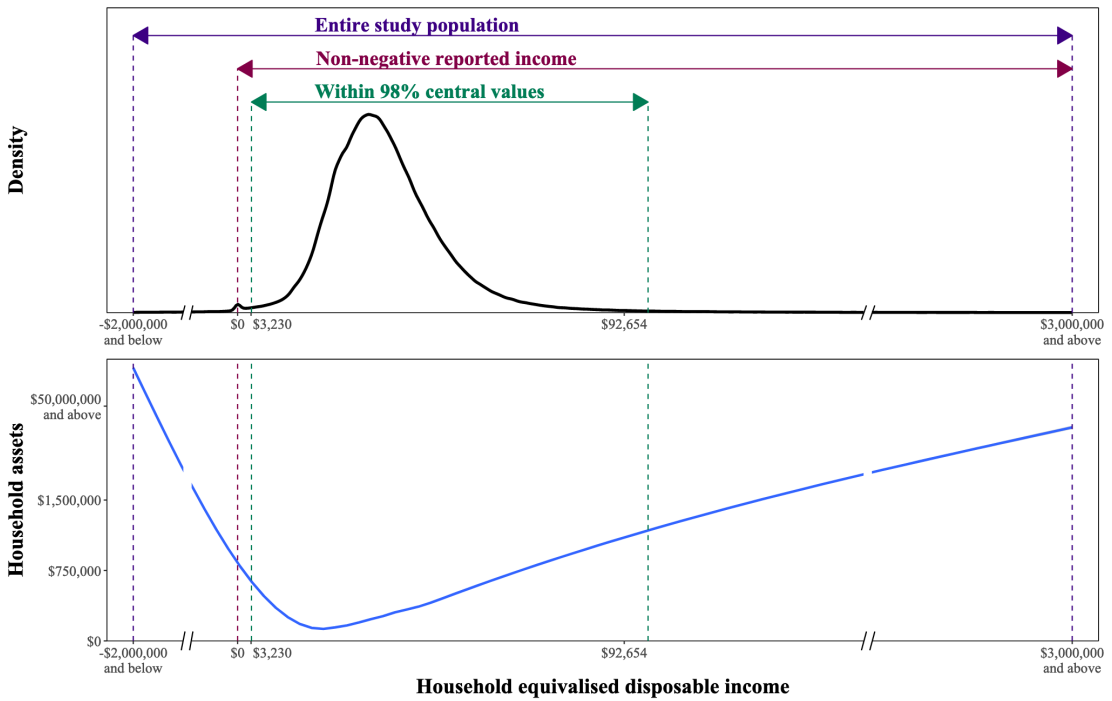
(a) Raw distribution of z-statistics.

Every test statistic published in the *AER*, *JPE* and *QJE*, 2005–2011. Source: Brodeur, Lé, Sangnier & Zylberberg (2016), *Star Wars: The Empirics Strike Back*, AEJ: Applied 8(1), Figure 1(a).

Two phenomena in this plot:

- **Spiky bars:** authors round, reporting `z = 2.00` rather than `z = 1.9973` — information about the data generating process
- **The camel shape:** a dip just below 1.96 and a clump just above it — what does this tell us about which studies get published?

Example: income in the registers



- Middle 98%: **$3,230** to **$92,654**
- Full range: **–$2M** to **$3M** — heavily skewed
- Most negative income = most assets: capital and business losses, not poverty
- Dropping negatives or trimming outliers changes the study population

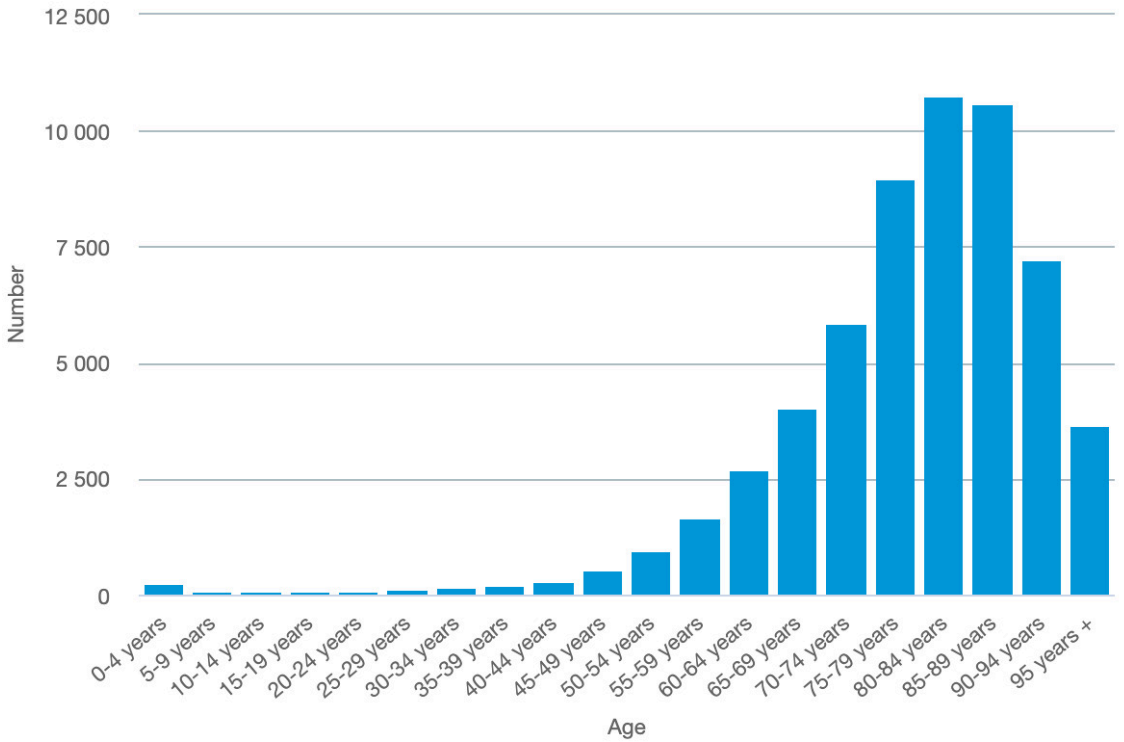
Source: Ejlskov & Plana-Ripoll (2025), *Income in epidemiological research*, J Epidemiol Community Health, Figure 1. Danish register data; caption cropped.

Continuous data: distribution plots

- Distributional plots are sometimes enough to convey a story/message
- Useful to build an intuition about the spread of data

Number of deaths disaggregated by age

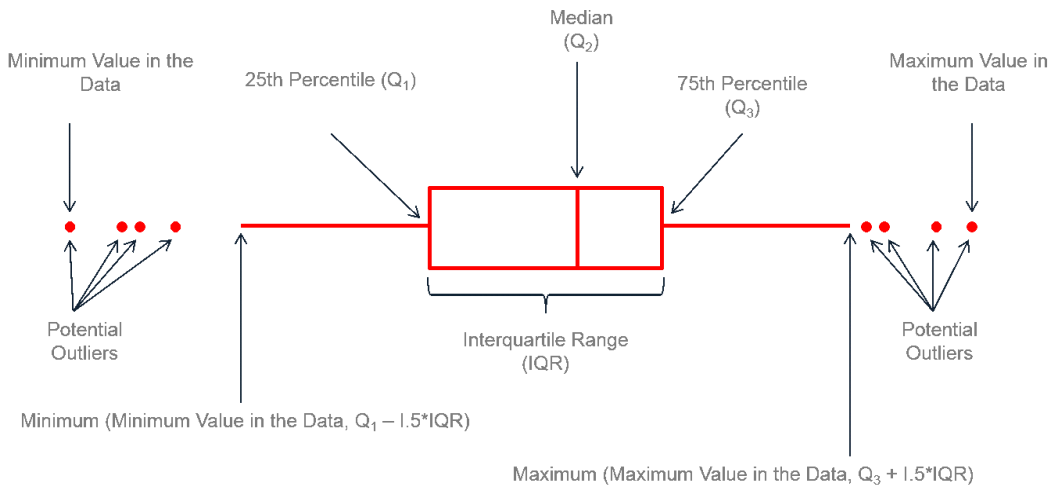
Time: 2025:



Continuous data: histograms

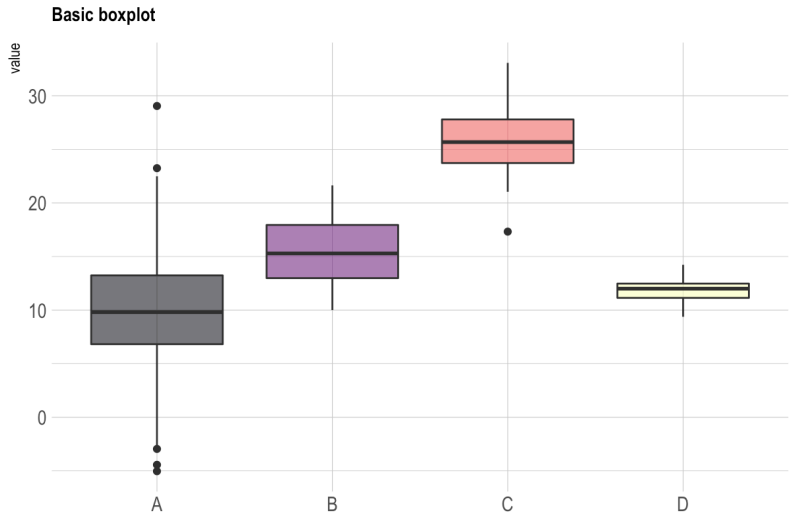
- Histogram / density plot: show the distribution of a variable of interest
- Example: age distribution of disease diagnoses, register data

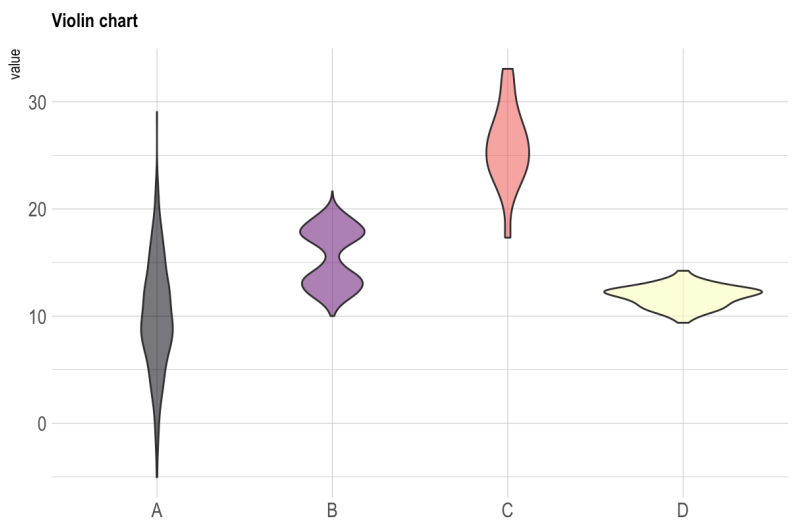
Continuous data: box plots

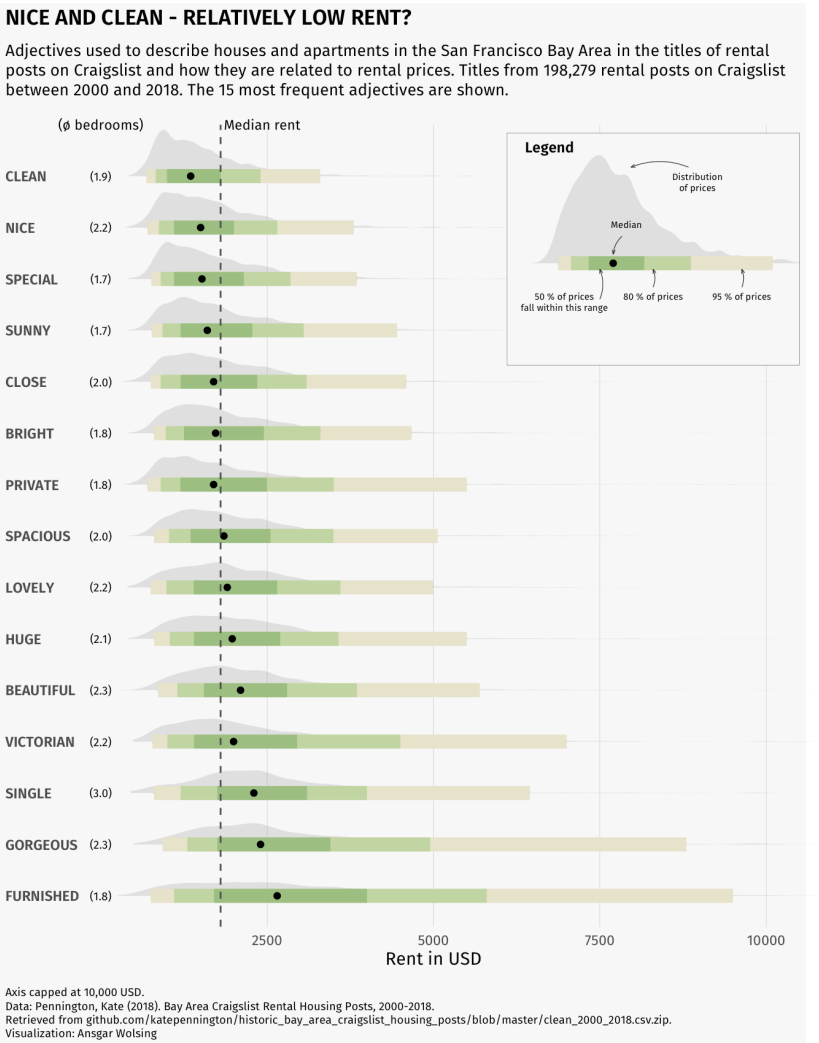


- Box plot: summarized plot which represents summary statistics in visual format
- Proceed with caution: information gets lost
- Best for comparing many distributions

Continuous data: box plot alternatives



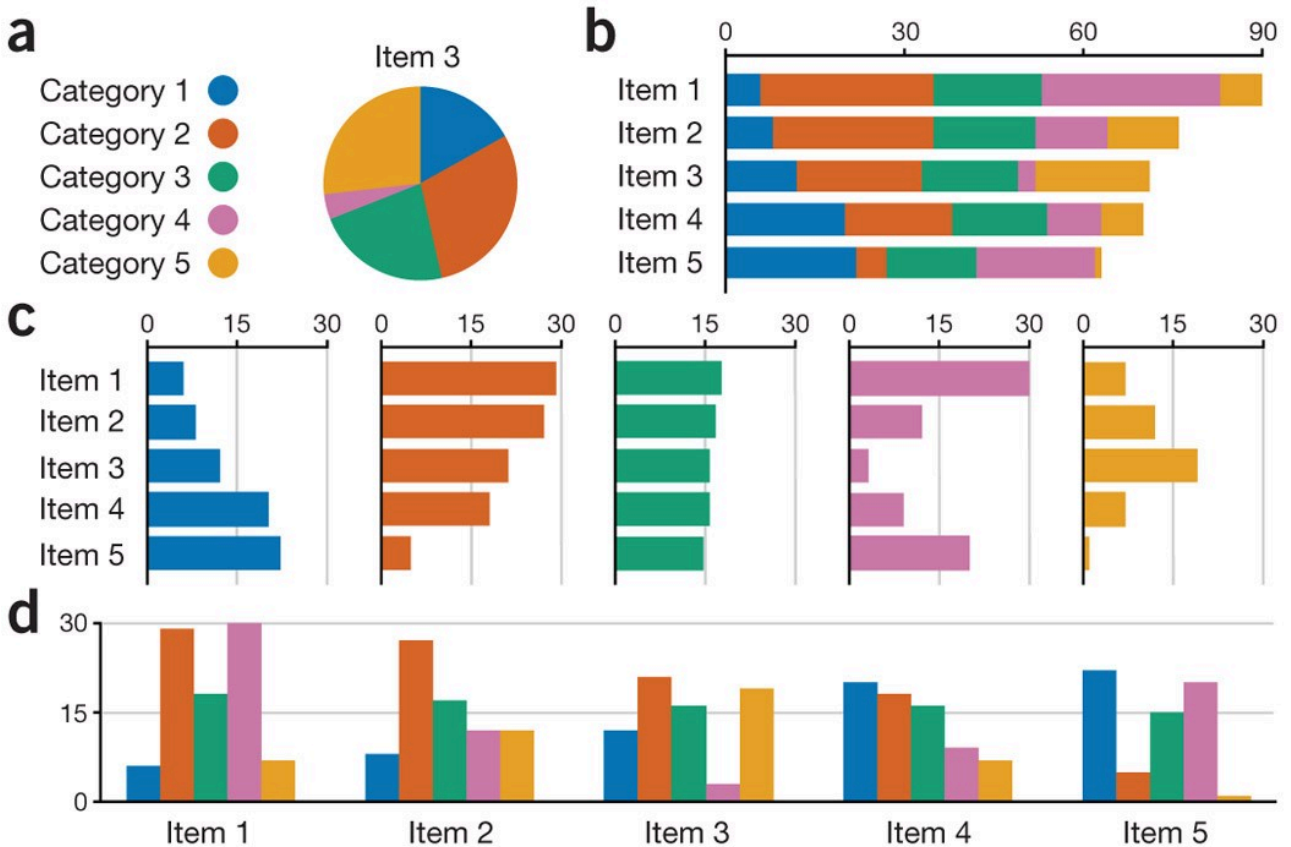




Ridge plot

Categorical data: bar plots

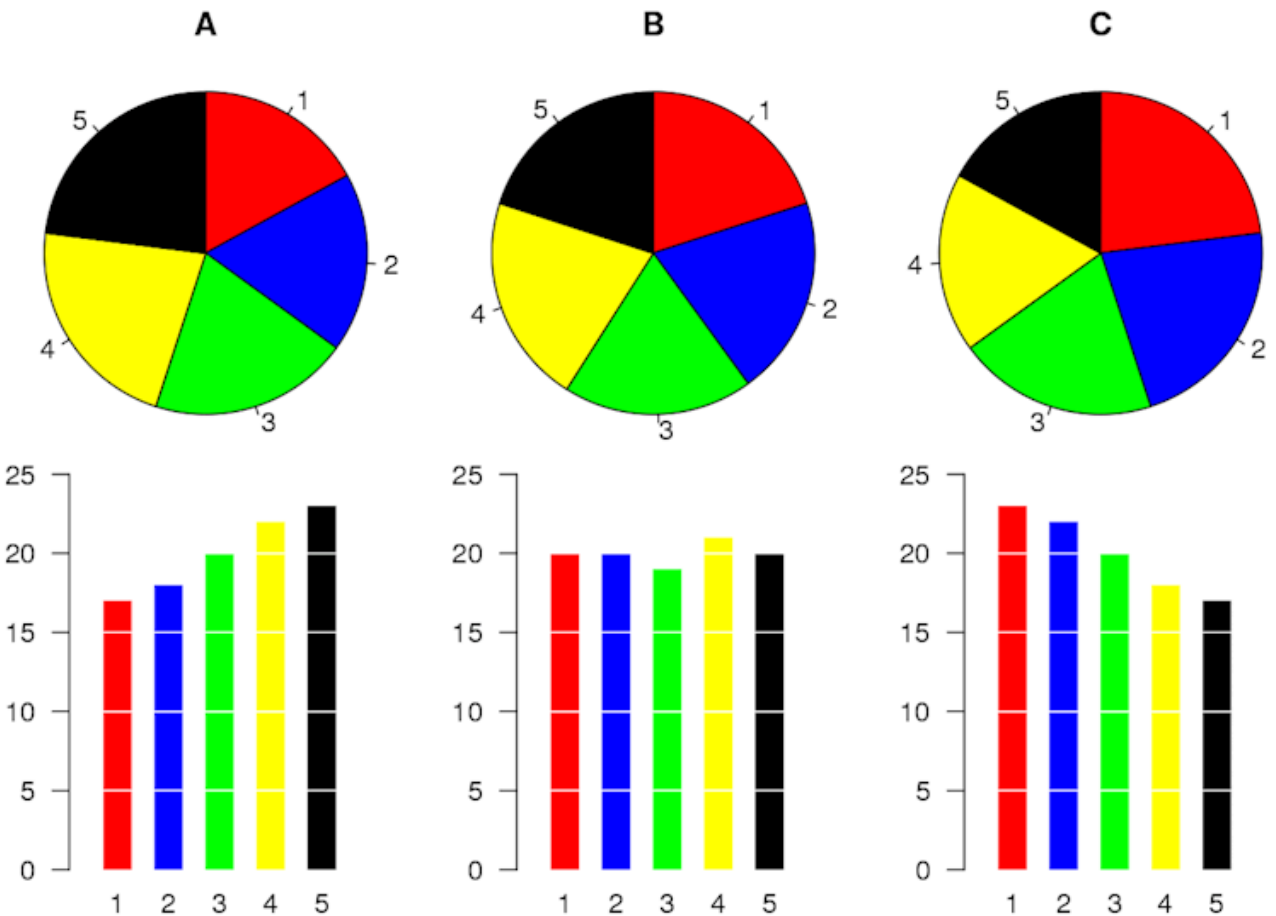
- Bar plot: represent frequency of category values
- Added complexity
  - Facets/groups
  - Position



https://www.nature.com/articles/nmeth.2807

Categorical data: pie charts

- Pie chart: represent categories as piece of a whole
- Issue: visual discernment



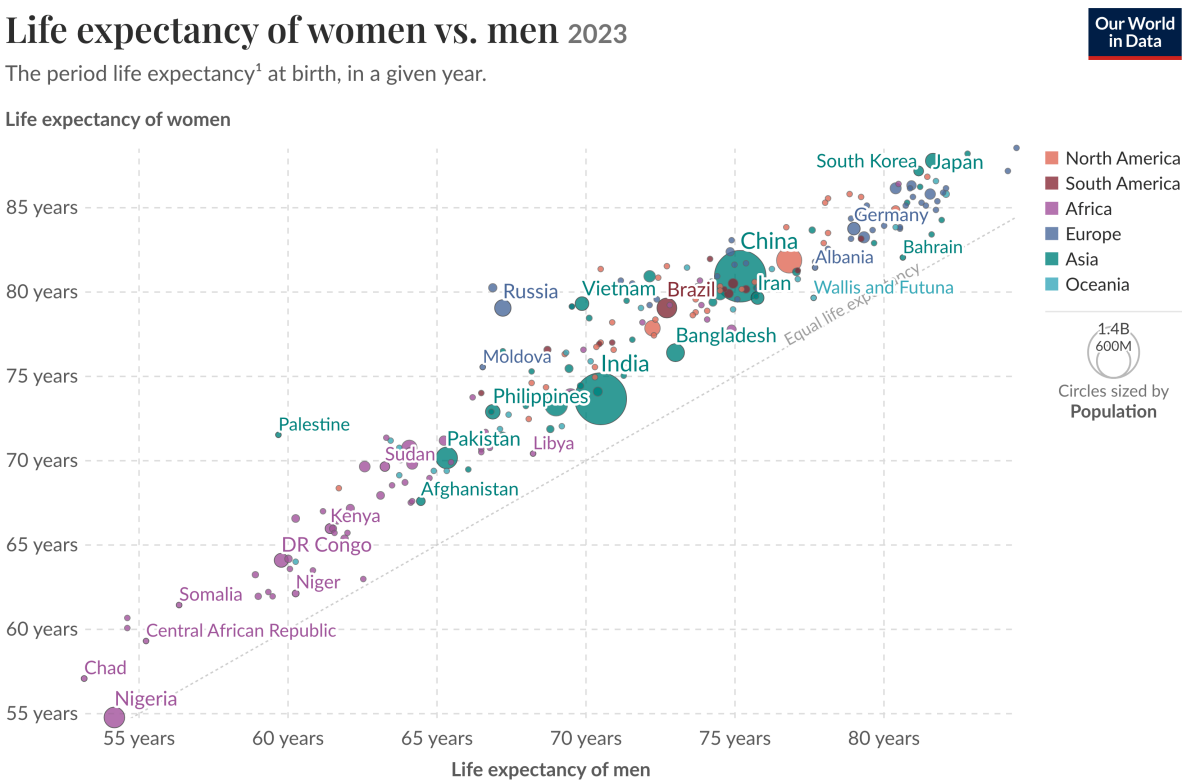
Multivariate EDA

Multivariate: comparisons and relationships

- Investigate comparisons or relationships between variables:
  - Comparison: sex differences and income
  - Relationships: income levels over time
- Visuals give more information than a correlation / coefficient
  - Therefore essential for good analysis

Multivariate: scatter plots

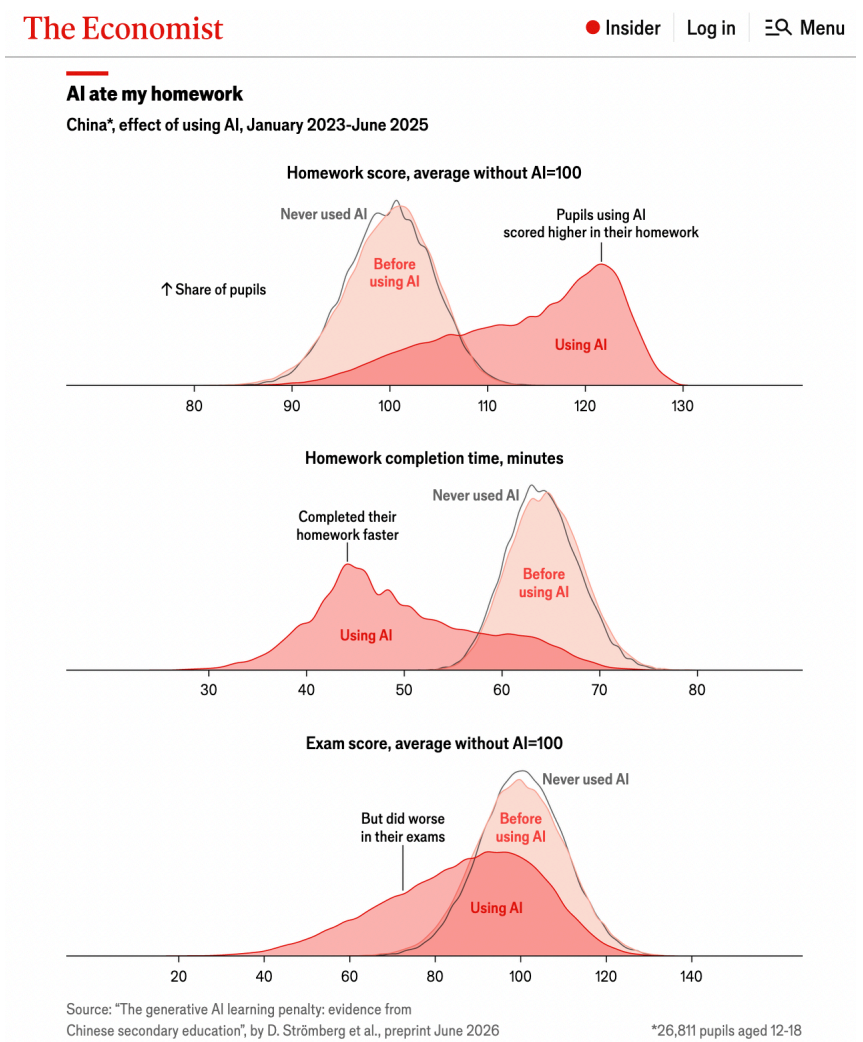
- Also useful to compare before & after steps in a data processing pipeline



1. Period life expectancy Period life expectancy is a metric that summarizes death rates across all age groups in one particular year.
For a given year, it represents the average lifespan for a hypothetical group of people, if they experienced the same age-specific death rates throughout their whole lives as the age-specific death rates seen in that particular year.
Learn more in our articles:
- Life expectancy – what does this actually mean?
- Period versus cohort measures: what's the difference?

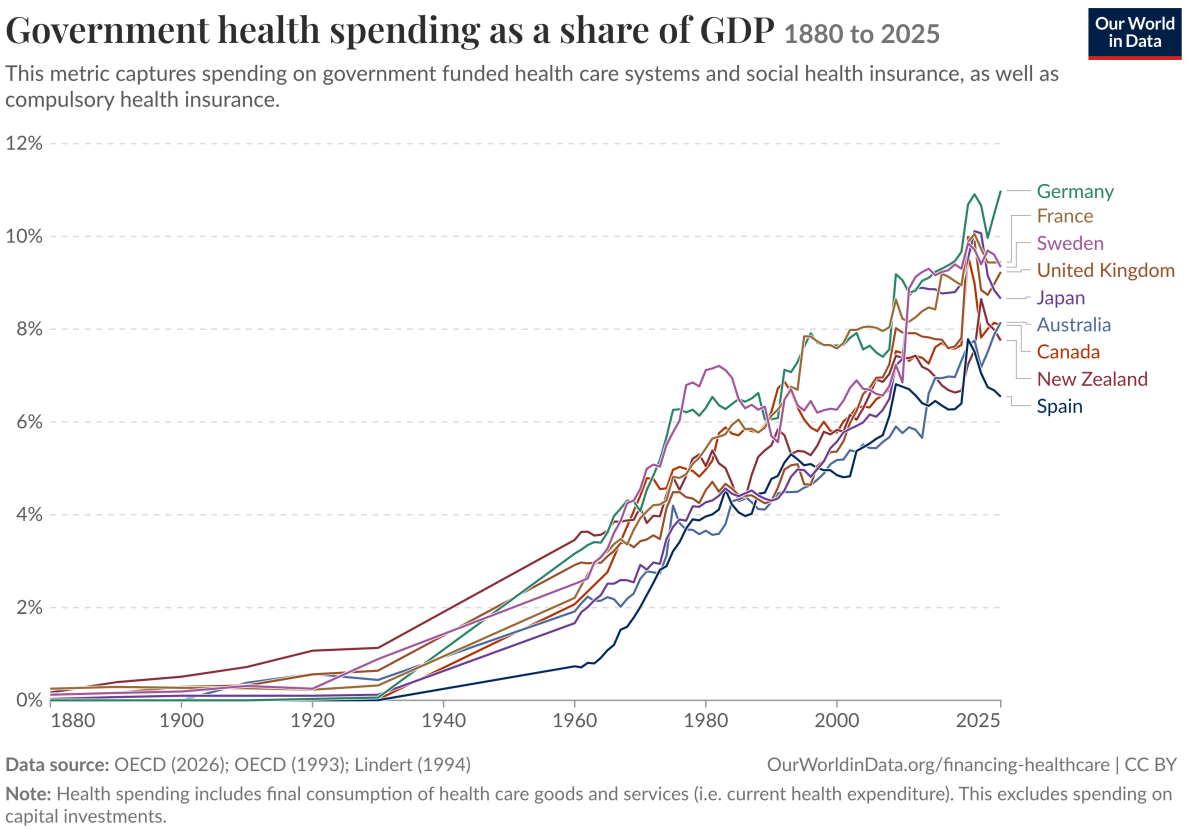
Multivariate: stratification and grouping

- Comparing distributions gives more information than summary statistics
- Consider overlays, panels depending on expected overlap



Time series & panel data

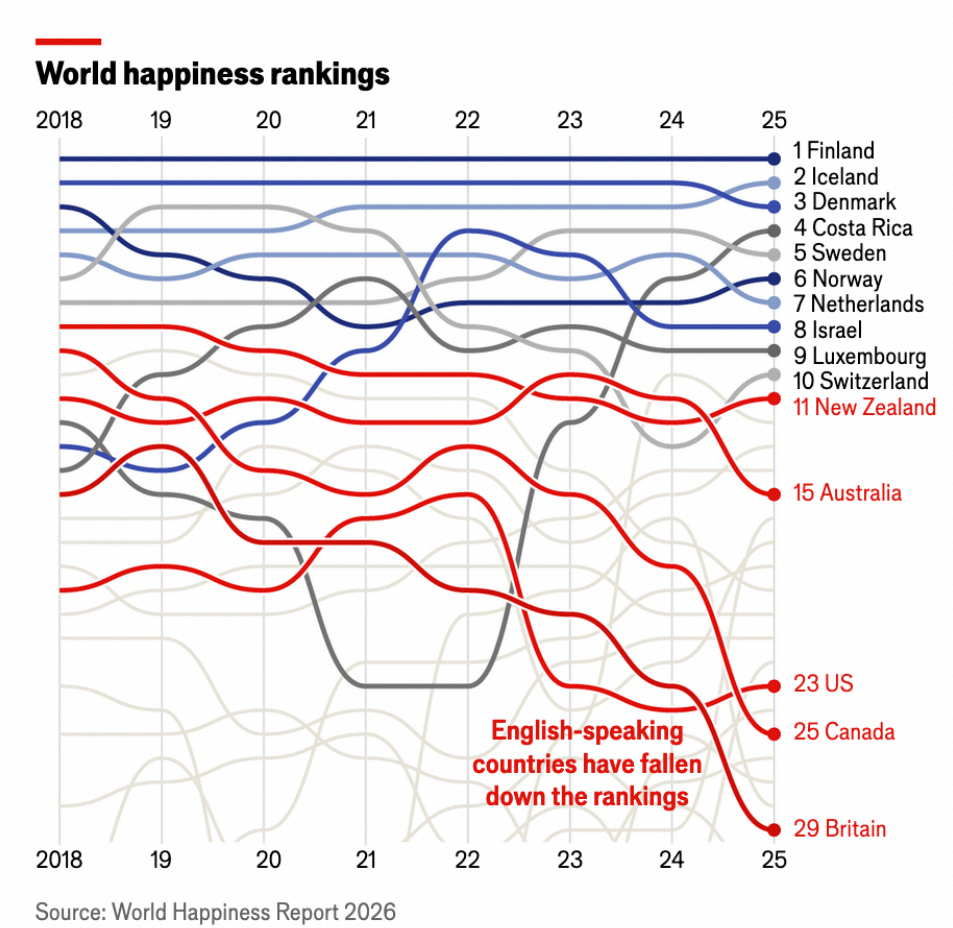
Time series: line plots



- Show continuity over time (age, etc.)
- Also used for comparison like scatters

Time series: ranking plots

- Useful to show comparisons where a baseline changes over time
- Somewhat uncommon in literature



https://www.economist.com/graphic-detail/2026/03/19/the-anglosphere-is-increasingly-miserable